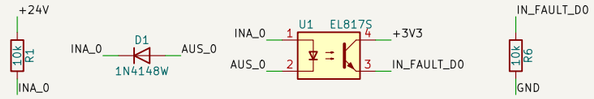
Trax HA – ZL 400 VF Board (XIAO) – Home Assistant controller for Dimplex ZL 300/400 VF (terminal board PCB3)

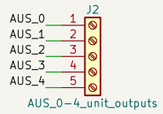
24 V SIDE (from the ventilation unit). Fully isolated from the ESP32 side: no shared ground. D7 and D8 protect against reversed 24 V.

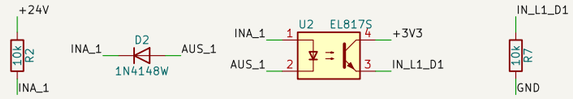


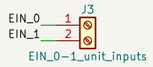
INPUTS: the unit pulls an AUS line to GND when active –> optocoupler LED lights (2.3 mA) –> GPIO goes HIGH (10k pull–down). Diode protects the optocoupler against reversed wiring.

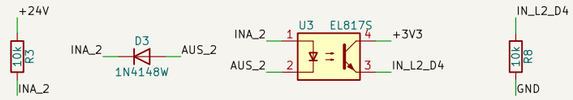


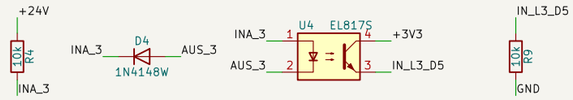






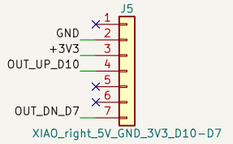




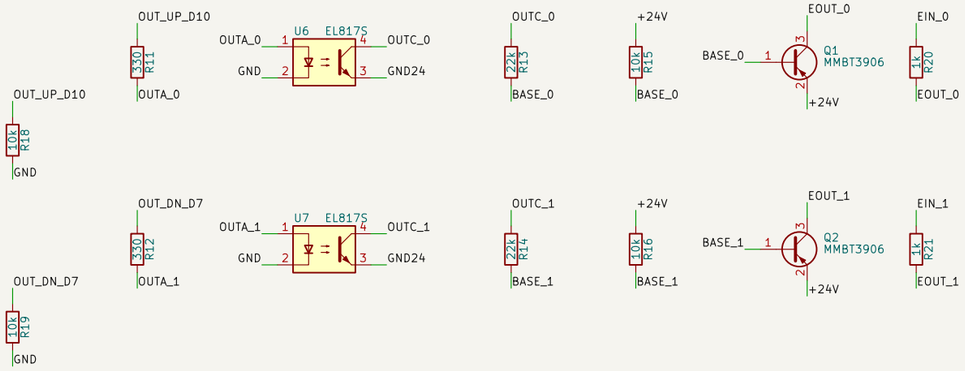


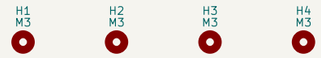
Seeed XIAO ESP32–C6 (ESP32–C5 fits with other GPIO numbers), USB–C power only.
Left header: D0 D1 D2 D3 D4 D5 D6 Right header: 5V GND 3V3 D10 D9 D8 D7
D2, D3, D8, D9 are left unused: they are boot strapping pins on some XIAO variants.





OUTPUTS: GPIO HIGH –> optocoupler –> PNP transistor puts +24 V on the EIN line (like pressing the panel button). R18/R19 hold the outputs off while the ESP32 boots or is unpowered. R20/R21 limit the current if an EIN line is shorted.





AUS_0 fault –> D0 AUS_1 level 1 –> D1 AUS_2 level 2 –> D4 AUS_3 level 3 –> D5 AUS_4 auto –> D6
EIN_0 up <– D10 EIN_1 down <– D7

Sheet: /	
File: ESP24V_HA_XIAO.kicad_sch	
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Size: A3	Date:
KiCad E.D.A. 10.0.7	**Rev:** Id: 1/1